A Dedicated Edge System and MCGI-YOLO for Fabric Defect Detection and Closed-Loop Control in Circular Knitting Machines

Kangzhao Tan[#], Jiarui Zhang[#], Junjie Yu, Heng Wu, Jing Lin[*], and Chunhua He[*]

***Abstract*—Defect detection is essential for circular knitting production, where persistent needle or yarn abnormalities lead to repeated fabric defects, resulting in severe material waste and economic losses. However, existing studies are generally not tailored to circular knitting machines and tend to be cost-prohibitive, which hinders their large-scale industrial deployment. To address this issue, we first design and implement a low-cost, dedicated defect detection and closed-loop control system for circular knitting machines. The system consists of a global-shutter camera, a Hall sensor, and a self-developed edge device. The Hall sensor signal triggers image acquisition and defect detection. When a defect is detected, the device stops the circular knitting machine to interrupt defect propagation and alerts the operator for corrective intervention. To further improve detection accuracy for diverse fabric defect morphologies, we propose a lightweight detector termed MCGI-YOLO. To model the directional continuity of horizontal and vertical elongated defects, morphology-aware context encoding (MCE) augments backbone features with spatially extended directional context. To accommodate variations in defect morphology and scale, gated interpolation upsampling (GIU) adapts the reconstruction of higher-level context before fusion with local features, using a compact content-dependent gate. On the test set of a self-collected dataset from a real circular-knitting production line, MCGI-YOLO achieves 97.2% mAP50 and 65.5% mAP50–95, improving on the YOLOv8n baseline by 6.7 and 7.1 percentage points, respectively, with a detection throughput of 22 FPS when deployed on the RK3566 edge device. These results demonstrate that the proposed system offers a cost-effective solution for large-scale deployment in real-time defect detection for circular-knitting production.**

***Index Terms*—Circular knitting machine, fabric defect detection, edge computing, MCE(morphology-aware context encoding), GIU(gated interpolation upsampling), YOLO.**

I. Introduction

CIRCULAR knitting machines produce weft-knit fabrics for garments such as T-shirts and sportswear [1]–[3]. Approximately 24,500 machines were shipped globally in 2025 [4]. With annual global textile waste estimated at 92 million tonnes, reducing avoidable production losses is important [5]. Conventional post-knitting inspection allows persistent needle or yarn abnormalities to generate repeated defects before discovery [6]–[8]. Real-time detection during knitting enables early intervention to limit this waste.

Edge computing supports such detection. Song et al. deployed EfficientDet-D0 on a Jetson TX2 for five fabric datasets covering dark red, grid, camouflage, digitally printed, and light blue fabrics [9]. Machado et al. used YOLO11n on a Jetson Orin Nano for solid-color cotton and linen fabrics on a rolling line [10]. Yan et al. deployed an improved YOLOv5s on an Ascend 310-based Atlas platform for warp-knitted lace on warp-knitting machines [11].

In other domains, edge implementations address rail defects on Jetson Xavier NX [12], insulator defects on Jetson TX2 NX [13], and glass defects on ZCU102 FPGA [14]. These general-purpose platforms require additional integration for the sensing, control, and installation needs of circular knitting machines. Their hardware cost and integration effort hinder deployment across large numbers of machines.

YOLO-based detectors combine efficient inference with defect localization [15]. For textile inspection, YOLOV4-TinyS combines lightweight convolutions, shallow-feature fusion, and attention for conveyor-based inspection [16]; AYOLOv7-tiny uses space-to-depth downsampling and hybrid attention for solid-color circular weft fabrics [17]. Improved YOLOv8 [18], FA-YOLO [19], and DGHR-YOLO [20] further explore attention, multiscale fusion, receptive-field enhancement, and deformable convolution. Related photovoltaic-cell and pipeline-weld studies develop task-specific feature pyramids, contextual relations, and lightweight rotated representations [21]–[23]. These advances address diverse inspection targets, but their accuracy–efficiency tradeoffs may not directly suit circular-knitting edge deployment. Horizontal and vertical elongated defects require extended structural context, while localized holes require local appearance evidence. Their varied morphologies and scales also challenge fixed feature reconstruction, and computationally intensive modules can limit throughput on low-cost hardware.

We therefore develop a dedicated detection and closed-loop control system integrating global-shutter imaging, SerDes

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transmission, rotation sensing, and an RK3566 edge device. Detection triggers machine stopping, defect upload, and operator notification. Within this system, the YOLOv8n-based MCGI-YOLO uses MCE to encode the morphology priors of elongated defects while retaining local C2f paths, and GIU to adapt feature reconstruction to variations in defect morphology and scale.

The main contributions are as follows.

(1) A dedicated, low-cost system integrates image acquisition, transmission, edge detection, and closed-loop machine control, with installation on a real circular-knitting production line.
(2) MCE incorporates the morphology priors of horizontal and vertical elongated defects through context modulation in a partial C2f branch, retaining local feature paths with small additional overhead.
(3) GIU adapts a shared spatial gate to blend two fixed interpolation paths, accommodating different defect morphologies and scales before fusion with lateral features.

II. Defect Detection and Closed-Loop Control System

A. System Overview

We design and implement a dedicated defect detection and closed-loop control system for circular knitting machines, as shown in Fig. 1. The system integrates a power adapter, a global-shutter camera with LED illumination, a SerDes image-transmission link, a Hall sensor, a self-developed edge device, and a cloud platform. The closed-loop workflow begins when the operator starts the machine. The Hall sensor detects machine rotation and triggers the edge device to activate the camera and LED illumination for image acquisition and defect detection. When a defect is detected, the edge device immediately sends a stop command to the machine controller to prevent further defect propagation, uploads the defect image and record to the cloud, and pushes an alert notification to the operator. After the operator inspects the reported defect, resolves the production fault, and restarts the machine, the rotation signal reactivates detection. This sequence forms a complete production-–detection-–control loop. The system is organized into four functional subsystems: power supply, image acquisition and transmission, and processor. The following subsections describe their designs and hardware implementation.

B. Power Supply Subsystem

To accommodate different industrial supply voltages across countries and regions, we design a power adapter with a 100–500 V AC input range and a 24 V DC output rated at 3 A. The adapter first supplies this output to the device. The device then supplies the LED illumination. By adjusting the current delivered to the LEDs, the device controls the illumination brightness to suit the image-acquisition conditions. This arrangement integrates LED power delivery and brightness control into the device.

C. Image Acquisition and Transmission Subsystem

The rapid rotation of circular knitting machines requires imaging that preserves defect geometry and fine spatial detail. We therefore select a SmartSens SC130GS global-shutter sensor(SmartSens Technology Co., Ltd., Shanghai, China) at 1280×1024 resolution, which exposes all pixels simultaneously to avoid motion-induced distortion from sequential row exposure. At a camera-to-fabric distance of approximately 10 cm, the field of view is approximately 16×12 cm, with the 16 cm dimension aligned with the horizontal direction of fabric motion.

Factory machinery creates an electromagnetically noisy environment, and installation requires the camera and edge device to be separated by a cable run. We therefore use serializer/deserializer (SerDes) transmission over shielded coaxial cable to accommodate remote camera placement and the interference environment. Specifically, the THCV241A serializer and THCV242A deserializer(THine Electronics, Inc., Tokyo, Japan) bridge the camera-side and processor-side MIPI CSI-2 interfaces. The devices support a maximum serial line rate of 4.0 Gb/s per lane, corresponding to a rated effective rate of 3.2 Gb/s [24], [25].

The camera is configured to transmit 1280×1024 RAW10 images at 30 frames/s. With $b = 10$ bits per pixel, the active-image payload rate is

$$\begin{aligned} R_{\mathrm{img}} &= WHbf_{\mathrm{cam}} \\ &= 1280 \times 1024 \times 10 \times 30 \\ &= 393.216\ \mathrm{Mb/s}, \end{aligned} \tag{1}$$

where W and H denote the image width and height, and f_{cam} is the camera acquisition rate. This corresponds to 49.152 MB/s, accounting for only 12.3% of the rated single-lane effective capacity, which provides a stable image input for the system's defect detection.

D. Processor Subsystem

The processor is selected according to the detection rate required by fabric motion and the camera field of view. Taking a typical large circular knitting machine as an example, with a machine diameter of 40 in ($D = 1.016$ m) and a rotation period of $T = 2$ s, the fabric travels approximately $\pi D = 3.192$ m per revolution. With an effective field-of-view width of $w = 0.16$ m along the direction of motion, the required frame rate is

$$f_{\min} = \frac{\pi D}{Tw} = \frac{\pi \times 1.016}{2 \times 0.16} \approx 9.97\ \mathrm{frames/s}. \tag{2}$$

We set the effective detection-throughput requirement to 10 frames/s. For this throughput target, we select the Rockchip RK3566(Rockchip Electronics Co., Ltd., Fuzhou, China) to balance processing capability, camera connectivity, and hardware cost. It integrates a quad-core Cortex-A55 CPU, an NPU rated at 1 TOPS for INT8 operations [26].

Table I summarizes the hardware specifications of platforms used in related defect detection studies. While high-performance platforms like the NVIDIA Jetson Orin Nano and Xilinx ZCU102 offer substantial computational power, their costs range from hundreds to over 3,000 USD, which

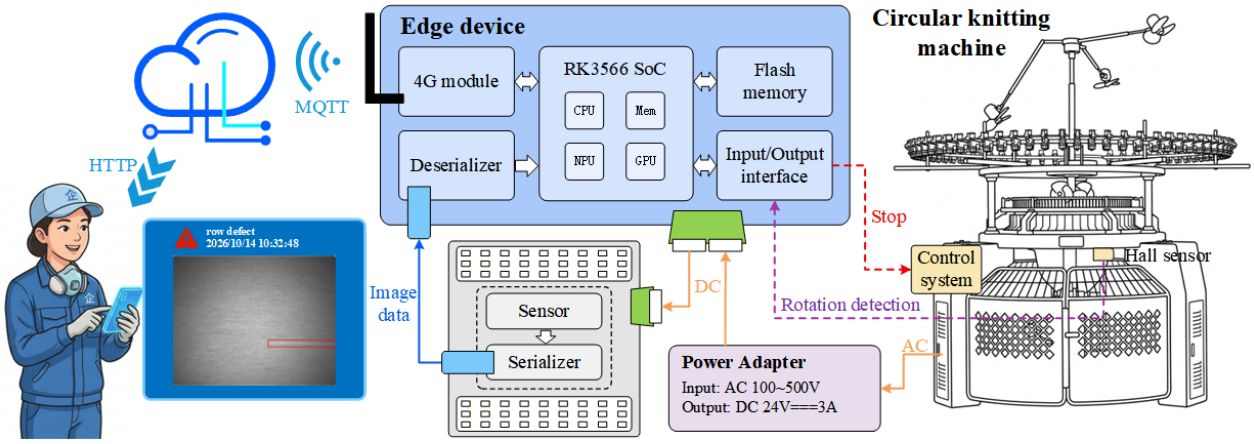


Fig. 1. System overview and subsystem interconnections. The power adapter supplies the device, which powers the LED illumination and controls its brightness by adjusting the output current. Images are transmitted through the serializer, coaxial link, and deserializer to the RK3566 processor. The 4G network connects the device to the cloud for monitoring and defect alerts. Wired input/output interfaces receive the Hall-sensor rotation signal and send stop commands to the machine controller.

TABLE I
HARDWARE PLATFORMS USED IN RELATED DEFECT DETECTION STUDIES.

Platform	Memory (GB)	Compute capability	Price (USD)
NVIDIA Jetson TX2 [9]	8	1.33 TFLOPS (FP16)	599
NVIDIA Jetson TX2 NX [13]	4	1.33 TFLOPS (FP16)	249
NVIDIA Jetson Xavier NX [12]	8	21 TOPS (INT8)	399
NVIDIA Jetson Orin Nano [10]	8	40 TOPS (INT8)	499
Huawei Ascend 310 [11]	8	16 TOPS (INT8)	550
AMD Xilinx ZCU102 [14]	4.5†	3.45 TOPS (INT8)‡	3234
Self-developed Rockchip RK3566 device(Ours)	4	1 TOPS (INT8)	60

†ZCU102 memory is the sum of 4 GB PS DDR4 and 0.5 GB PL DDR4. ‡The 3.45 TOPS value is AMD's reference INT8 peak for three B4096 DPU cores.

can be prohibitive for large-scale deployment. In contrast, the Rockchip RK3566 provides a cost-effective hardware basis for the proposed system, despite its rated compute capability of only 1 TOPS (INT8). Most notably, at a hardware cost of only 60 USD, our solution offers a significant cost advantage over comparable industrial platforms, making it highly suitable for large-scale deployment in textile factories.

E. Hardware Implementation

Based on the design requirements of the subsystems described above, we develop the corresponding hardware shown in Fig. 2. The camera assembly (A) combines the image sensor and LED illumination. We design a camera-side interface board (B) that integrates the serializer and coaxial connector, and a mainboard (C) built around the RK3566. The mainboard integrates memory and flash storage, the camera deserializer and coaxial connector, a 4G module with SIM-card and antenna interface, input/output interface, and power-management circuitry. This board-level integration brings image reception, local inference, network communication, machine control, and illumination control into a dedicated edge device.

Fig. 3 shows the system installed on a circular knitting machine in a real production line. The highlighted components include the start button, Hall sensor, industrial camera, and RK3566 edge device.

III. DETECTION ALGORITHM

A. MCGI-YOLO Architecture

Fig. 4 presents MCGI-YOLO, which is built upon YOLOv8n [27]. MCE is embedded in the backbone C2f blocks at P3 (stride 8) and P4 (stride 16). It introduces horizontal and vertical context to model the directional continuity of elongated defects while preserving the original local feature paths.

GIU replaces the two nearest-neighbor upsampling operations in the top-down neck. It adapts the balance between response replication and smoothing to accommodate variations in defect morphology and scale before fusion with lateral backbone features. The remaining backbone stages, bottom-up aggregation path, and three-scale detection head are retained [28], [29].

B. Morphology-Aware Context Encoding

MCE encodes the continuity of horizontal and vertical elongated defects through spatially extended context while retaining local C2f processing (Fig. 5). The categories describe dominant fabric directions and allow small angular deviations. Row and column pooling follows the directional encoding principle of coordinate attention [30].

1) Partial-Channel Placement: The first C2f 1×1 convolution splits its output into branches Z_0 and Z_1, each with $c = \lfloor C_{\text{out}}/2 \rfloor$ channels. The original bottlenecks sequentially generate $Z_{i+1} = \mathcal{B}_i(Z_i)$, $i = 1, \ldots, n$. MCE acts only on

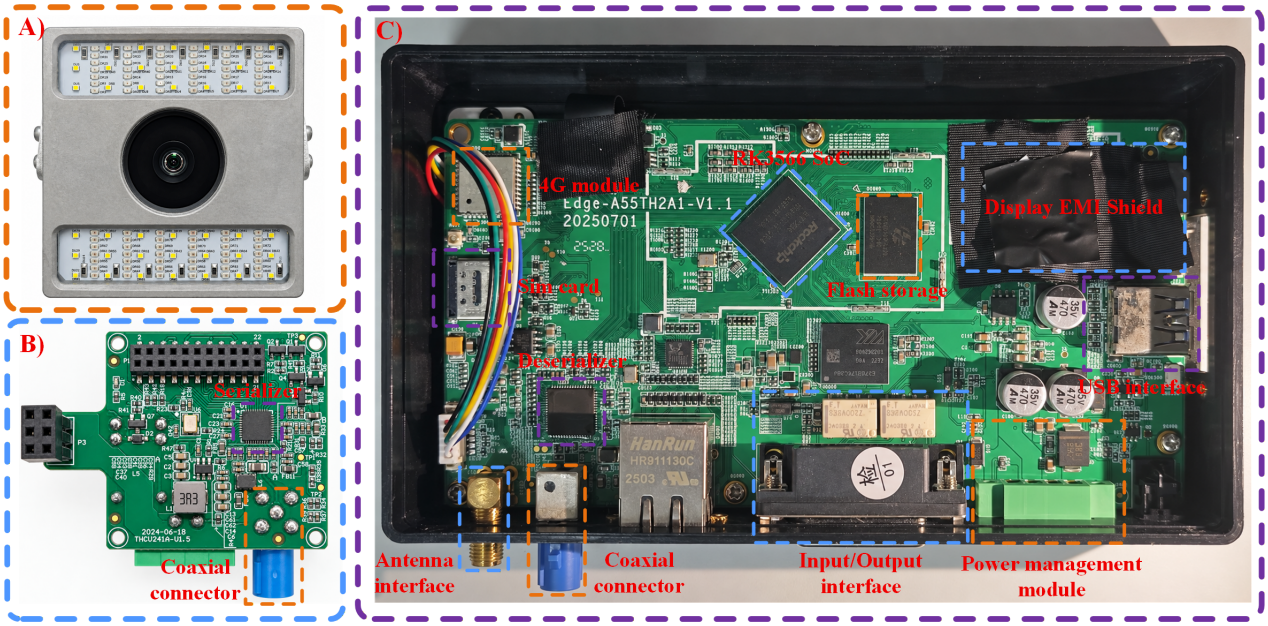


Fig. 2. Hardware of the defect detection system: (A) industrial camera with LED illumination; (B) camera-side interface board with a serializer and coaxial connector; and (C) RK3566 edge board with the 4G communication, flash storage, camera-interface, input/output interface, and power-management components.

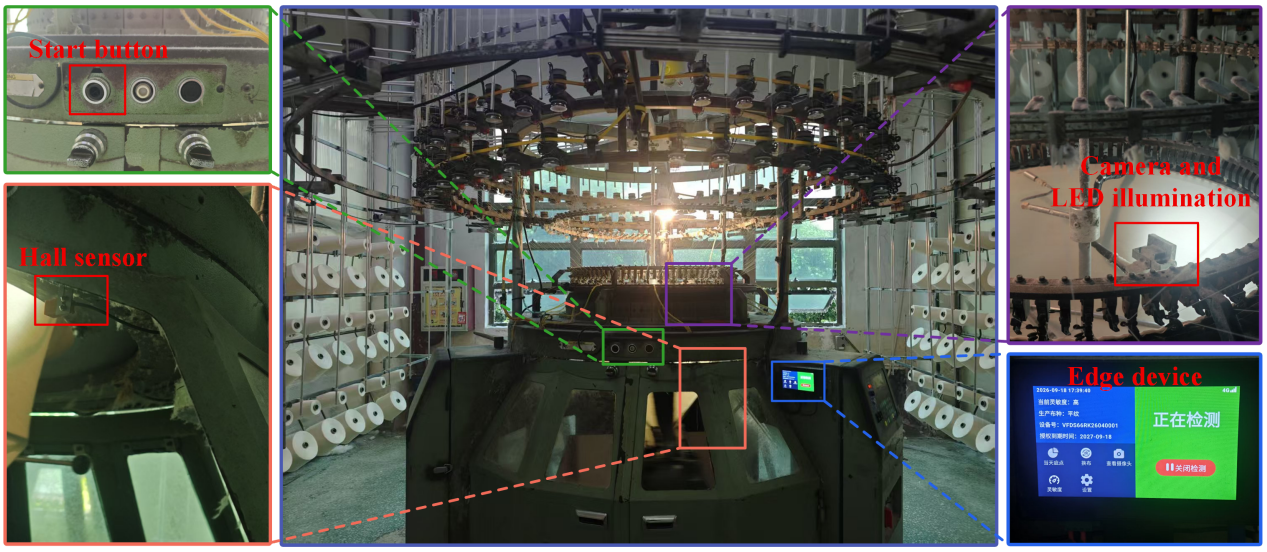


Fig. 3. Actual deployment on the circular knitting machine. The insets highlight the start button, Hall sensor, industrial camera, and RK3566 edge device, respectively.

$F = Z_{n+1}$, split into active channels $F_a \in \mathbb{R}^{q\times H\times W}$ and bypass channels $F_b \in \mathbb{R}^{(c-q)\times H\times W}$, where $q = \max(1, \lfloor 0.25c \rfloor)$. The one-quarter ratio thus applies to the final hidden branch.

2) Spatial Context Aggregation: For active channel a, the row and column descriptors are

$$R_h(a,h,1) = \frac{1}{W}\sum_{w=1}^{W} F_a(a,h,w), \tag{3}$$

$$R_w(a,1,w) = \frac{1}{H}\sum_{h=1}^{H} F_a(a,h,w). \tag{4}$$

The descriptors retain row and column positions, respectively, and are filtered and combined as

$$T = \mathrm{DW}_{3\times1}(R_h) + \mathrm{DW}_{1\times3}(R_w). \tag{5}$$

The biased depthwise filters preserve descriptor dimensions through padding, and spatial broadcasting produces $T \in \mathbb{R}^{q\times H\times W}$. Full-axis pooling supplies the long-range context.

3) Residual Modulation and Feature Reuse: The signed context map modulates the active channels:

$$A = 2\,\mathrm{HS}(T) - 1, \tag{6}$$

$$F_a' = F_a + \gamma \odot F_a \odot A, \tag{7}$$

where HS is Hardsigmoid, $\odot$ denotes element-wise multiplication with broadcasting, and $\gamma \in \mathbb{R}^{q\times1\times1}$ is an unconstrained

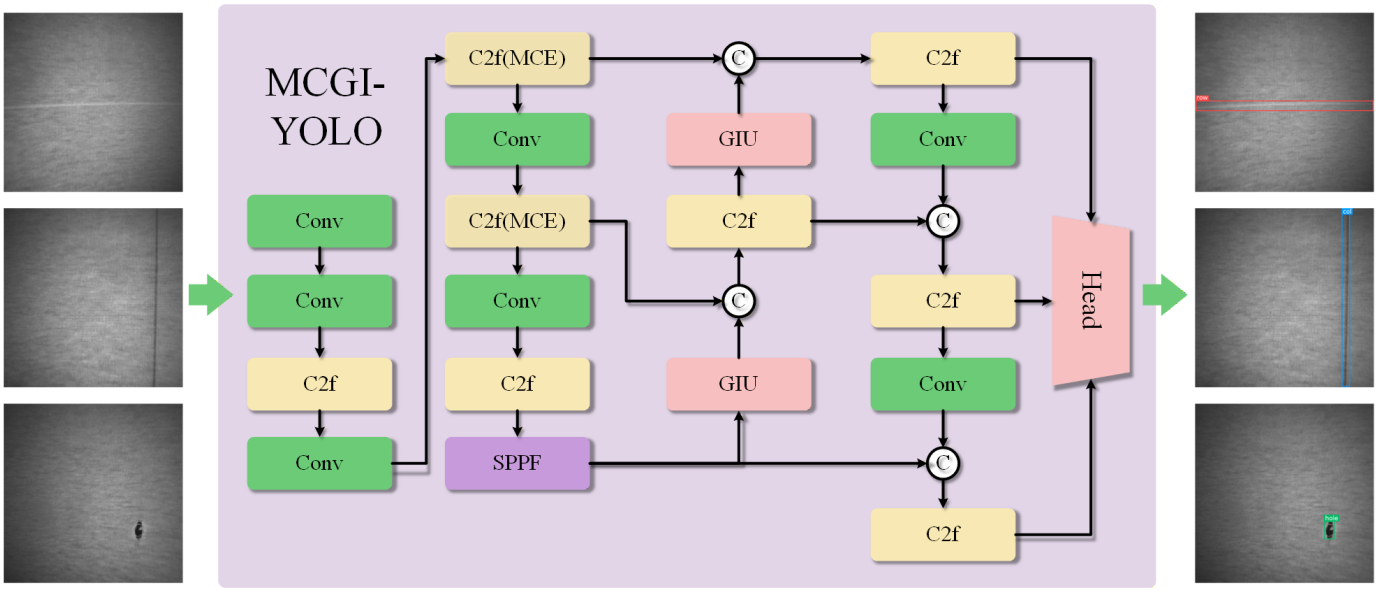


Fig. 4. Architecture of MCGI-YOLO. MCE operates within the stride-8 and stride-16 backbone C2f stages. GIU replaces the two top-down upsampling operations. The local bottlenecks, bottom-up aggregation path, and three-scale detection head are retained from YOLOv8n.

learnable scale. The bypass channels and earlier C2f branches are retained:

$$Y = \phi_2\big([Z_0, \ldots, Z_n, [F'_a, F_b]]\big), \tag{8}$$

where ϕ_2 is the original final 1×1 convolution with batch normalization and SiLU. Initializing $\gamma = 0$ preserves the corresponding C2f mapping when shared weights are identical. The added filters and scales contain $9q$ parameters; pooling and modulation also process the spatial feature grid.

C. Gated Interpolation Upsampling

Elongated defects and holes of different sizes place different demands on continuity, response concentration, and boundary representation. GIU adapts the reconstruction of higher-level context before fusion with lateral features F_3 and F_4, which retain local texture and localization cues (Fig. 5). Complementary gating has precedents in highway networks [31]; here, a shared spatial gate blends two fixed interpolation paths.

1) Two Fixed Interpolation Paths: For $X \in \mathbb{R}^{C\times H\times W}$, nearest-neighbor and bilinear upsampling produce

$$N = \mathrm{Nearest}_2(X), \quad B = \mathrm{Bilinear}_2(X), \tag{9}$$

where $N, B \in \mathbb{R}^{C\times 2H\times 2W}$ and bilinear interpolation uses `align_corners=False`.

2) Shared Spatial Gate: A gate is predicted at the input resolution and upsampled:

$$G = \mathrm{Nearest}_2\big(\mathrm{HS}(W_g * X + b_g)\big), \tag{10}$$

where $*$ denotes convolution, $W_g \in \mathbb{R}^{1\times C\times 3\times 3}$, and b_g is a scalar bias. Padding of one preserves spatial dimensions. The single-channel gate $G \in [0,1]^{1\times 2H\times 2W}$ is broadcast across feature channels.

3) Gated Reconstruction: The output is a convex combination:

$$Y = (1 - G) \odot N + G \odot B. \tag{11}$$

A gate near zero favors response replication, while a gate near one favors bilinear smoothing. Initializing $W_g = 0$ and $b_g = -2.7$ gives

$$G_{\text{init}} = 0.05, \qquad Y_{\text{init}} = 0.95N + 0.05B. \tag{12}$$

Each GIU adds $9C + 1$ parameters, with $9CHW$ multiply–accumulate operations for the gate convolution. Interpolation and mixing add operations and memory traffic, so deployment speed is measured on the target device. The individual and combined module effects are evaluated in Section V-B.

IV. Experimental Setup

A. Dataset

The dataset was collected from a real circular-knitting production line and covers three defect categories: horizontal defects(*row*), vertical defects(*col*), and holes(*hole*). (*row*). It comprises 403 images with 468 annotated defect instances, divided into training, validation, and test sets of 282, 40, and 81 images, respectively, as summarized in Table II.

TABLE II
IMAGE-LEVEL SPLIT AND DEFECT-INSTANCE COUNTS IN THE REAL PRODUCTION-LINE DATASET.

Split	Images	Instances	Row	Col	Hole
Train	282	329	174	75	80
Validation	40	48	24	13	11
Test	81	91	60	12	19

B. Implementation Details

Table III summarizes the training and deployment settings. MCGI-YOLO and its component-ablation variants share the same `yolov8n.pt` initialization as the YOLOv8n baseline, and all component comparisons use a common training protocol with matched random seeds. Other comparison models use

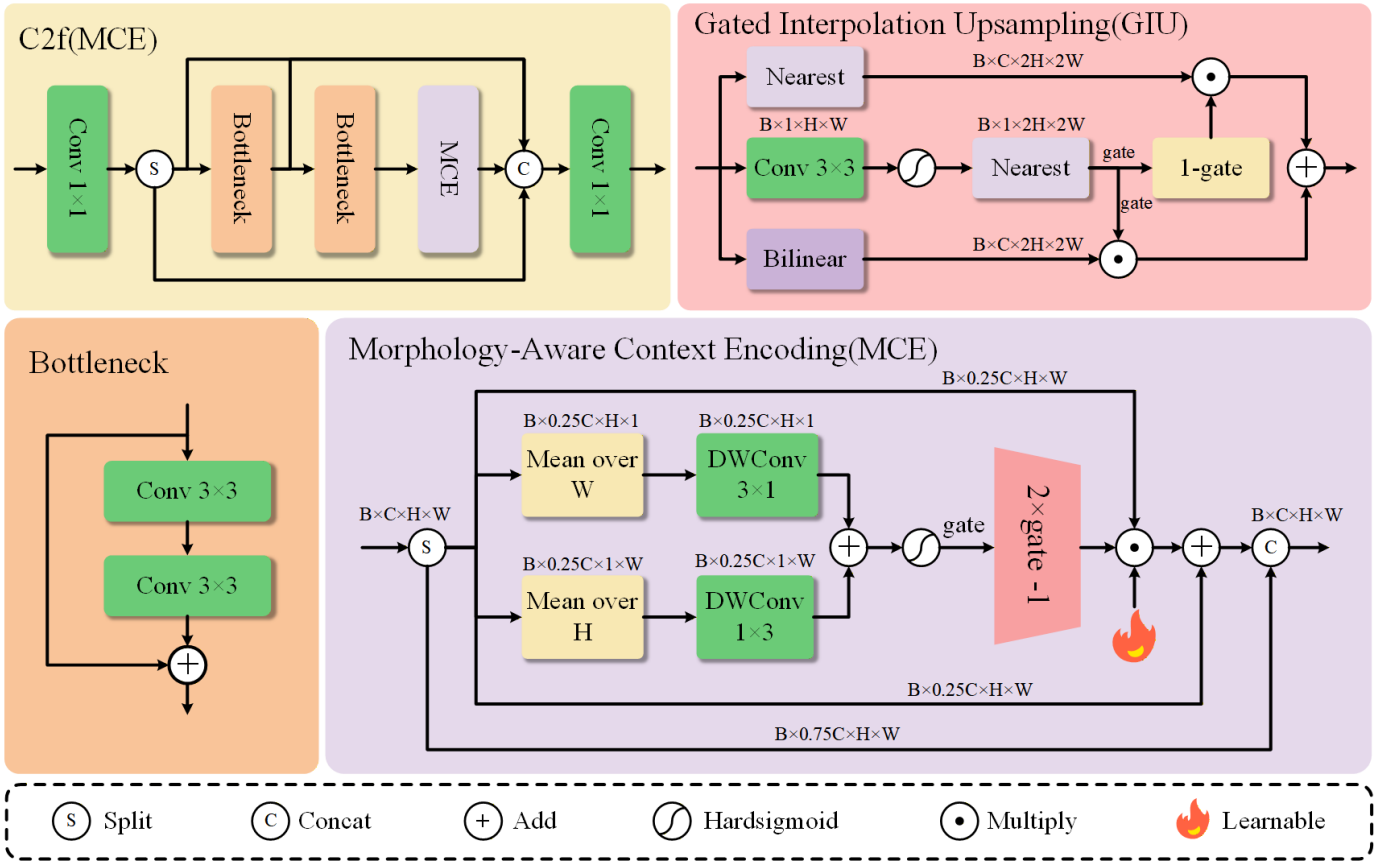


Fig. 5. Structures of C2f with MCE and GIU. MCE embeds directional context modulation in a partial C2f branch while retaining local feature paths. GIU blends nearest-neighbor and bilinear interpolation using a spatial gate shared across channels.

TABLE III
IMPLEMENTATION PROTOCOL. TRAINING SETTINGS DESCRIBE THE ULTRALYTICS EXPERIMENTS; DEPLOYMENT SETTINGS DESCRIBE THE INSTALLED EDGE DEVICE.

Category	Parameter	Specification
Training setup	Hardware	Intel i7-13700KF; NVIDIA RTX 3090 (24 GB)
	Software	Ubuntu 24.04, Python 3.10.20, PyTorch 2.4.1+cu124, and Ultralytics 8.4.140
Deployment setup	Hardware	Rockchip RK3566
	System	Android 13
	Quantization	INT8
Data protocol	Input resolution	Fixed at 640×640
Training dynamics	Batch	16
	Maximum epochs	300
	Patience	80 epochs
	Optimizer	AdamW

available pretrained checkpoints from their respective implementations.

The RT-DETR-R18 entry uses the RT-DETRv2 implementation from the official repository [32].

C. Evaluation Metrics

Intersection over union (IoU) measures the overlap between predicted and ground-truth bounding boxes as the ratio of their intersection area to their union area. Detection accuracy is evaluated using mAP50 and mAP50–95. Let $a_c(t)$ denote the average precision for class c at IoU threshold t. The metrics are computed as

$$\mathrm{mAP}_{50} = \frac{1}{N_c}\sum_{c=1}^{N_c} a_c(0.50), \tag{13}$$

$$\mathrm{mAP}_{50:95} = \frac{1}{10N_c}\sum_{c=1}^{N_c}\sum_{t\in T} a_c(t), \tag{14}$$

where $N_c = 3$ and $T = \{0.50, 0.55, \ldots, 0.95\}$. Both metrics are reported as percentages, with improvements expressed in percentage points.

Model parameters and FLOPs are measured at 640×640 input resolution using each model's respective framework. For FPS comparison, all models are evaluated on an RTX 3090 with FP32 precision, batch size 1, and 640×640 inputs. This measurement is separate from the RK3566 deployment result.

V. RESULTS AND DISCUSSION

A. Comparison With Other Detectors

Table IV compares the test-set accuracy, model complexity, and inference throughput of the evaluated detectors. MCGI-YOLO achieves the highest mAP50 of 97.2% and the second-highest mAP50–95 of 65.5%. Compared with the YOLOv8n baseline, these scores increase by 6.7 and 7.1 percentage points, respectively. The parameter count increases from 3.01 M to

TABLE IV
DETECTOR COMPARISON ON THE TEST SET. FPS DENOTES FP32 MODEL-FORWARD THROUGHPUT ON AN RTX 3090 WITH BATCH SIZE 1 AND A 640×640 INPUT. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY.

Model	Params (M)	FLOPs (G)	mAP50 (%)	mAP50–95 (%)	FPS
YOLOv9t [33]	**2.12**	8.5	90.1	53.3	291
YOLOv10n [34]	2.78	8.8	88.2	58.5	<u>524</u>
YOLO11n [27]	2.62	<u>6.7</u>	93.3	60.8	463
YOLO12n [35]	2.60	7.6	89.0	56.8	326
YOLO26n [27]	<u>2.57</u>	**6.2**	90.5	58.2	425
RT-DETR-R18 [36]	20.09	60.7	95.2	**67.7**	146
DETR-R50 [37]	41.28	77.3	96.4	53.6	101
Deformable DETR-R50 [38]	39.8	158.7	<u>96.8</u>	64.3	49
Faster R-CNN-R50 [39]	41.1	181.5	96.1	64.7	66
YOLOv8n (Baseline) [27]	3.01	8.1	90.5	58.4	**554**
MCGI-YOLO	3.02	8.2	**97.2**	<u>65.5</u>	486

3.02 M, and FLOPs increase from 8.1 G to 8.2 G. Model-forward throughput decreases from 554 to 486 FPS on the RTX 3090. Thus, the accuracy improvement comes with only a slight increase in model size and computation, at the cost of a moderate drop in throughput.

Among the other lightweight YOLO variants, YOLO11n achieves the highest accuracy, with 93.3% mAP50 and 60.8% mAP50–95. MCGI-YOLO improves these scores by 3.9 and 4.7 percentage points while achieving 486 FPS compared with 463 FPS for YOLO11n. Several YOLO variants have fewer parameters or lower FLOPs than MCGI-YOLO, but their accuracy is lower on this test set. These results demonstrate that adapting the model to this specific task yields better performance than using generic lightweight detectors.

RT-DETR-R18 obtains the highest mAP50–95 of 67.7%, exceeding MCGI-YOLO by 2.2 percentage points, whereas its mAP50 is 2.0 percentage points lower. This advantage at stricter matching thresholds comes with 20.09 M parameters and 60.7 G FLOPs, compared with 3.02 M and 8.2 G for MCGI-YOLO. Its throughput is also lower, at 146 FPS. MCGI-YOLO additionally outperforms DETR-R50, Deformable DETR-R50, and Faster R-CNN-R50 on both mAP metrics with fewer parameters and FLOPs. Overall, MCGI-YOLO provides a favorable accuracy–efficiency tradeoff among the evaluated configurations, although it does not achieve the highest mAP50–95 or the highest throughput.

Fig. 6 compares predictions for horizontal defects, vertical defects, and a hole. In the fourth row, YOLOv8n produces several boxes along a single vertical defect, while MCGI-YOLO produces one box spanning the annotated defect. In the final row, YOLOv8n divides the hole into multiple overlapping predictions, whereas MCGI-YOLO produces a single box that more closely follows the annotated extent. These examples illustrate differences in prediction fragmentation and box localization that complement the aggregate accuracy results.

B. Component Ablation

Table V defines four configurations: the original YOLOv8n baseline, MCE alone, GIU alone, and their combination. The MCE-only model retains nearest-neighbor upsampling, while the GIU-only model retains the original backbone C2f blocks.

TABLE V
COMPONENT ABLATION OF MCE AND GIU.

Variant	MCE	GIU	Params(M)	FLOPs(G)	mAP50	mAP50–95
YOLOv8n	–	–	3.01	8.10	90.5	58.4
+ MCE	✓	–	3.01	8.2	93.9	63.6
+ GIU	–	✓	3.01	8.2	94.4	63.0
MCGI-YOLO	✓	✓	3.02	8.2	97.2	65.5

The comparison isolates the two architectural changes under the common training protocol. The finalized results will report both individual effects and the joint configuration, together with variation across matched seeds and class-specific results for elongated defects and holes.

Feature visualizations in Fig. 7 complement the numerical analysis by comparing responses at corresponding feature stages on the same images. They are interpreted alongside localization and classification results rather than as a separate quantitative measure of detection performance.

C. Model Deployment and On-Device Performance

To deploy MCGI-YOLO on the RK3566 edge device, the trained model is exported to ONNX with a fixed input size of 640×640 and then converted into an INT8 RKNN model using RKNN-Toolkit2 [40]. For 640×640 inputs, the deployed INT8 model achieves a measured inference throughput of 22 FPS on the RK3566. This throughput is 2.2 times the 10 FPS design target derived in Section II-D, meeting the model-inference rate required for the specified machine speed and camera field of view.

VI. CONCLUSION

This paper presents a dedicated defect detection and closed-loop control system for circular knitting machines, integrating image acquisition, SerDes transmission, edge inference, machine stopping, and cloud alerts. The lightweight MCGI-YOLO combines MCE for elongated-defect morphology encoding with GIU for adaptive feature reconstruction. On the self-collected test set, it achieves 97.2% mAP50 and 65.5% mAP50–95, improving over YOLOv8n by 6.7 and 7.1 percentage points. INT8 deployment on the RK3566 achieves 22 FPS,

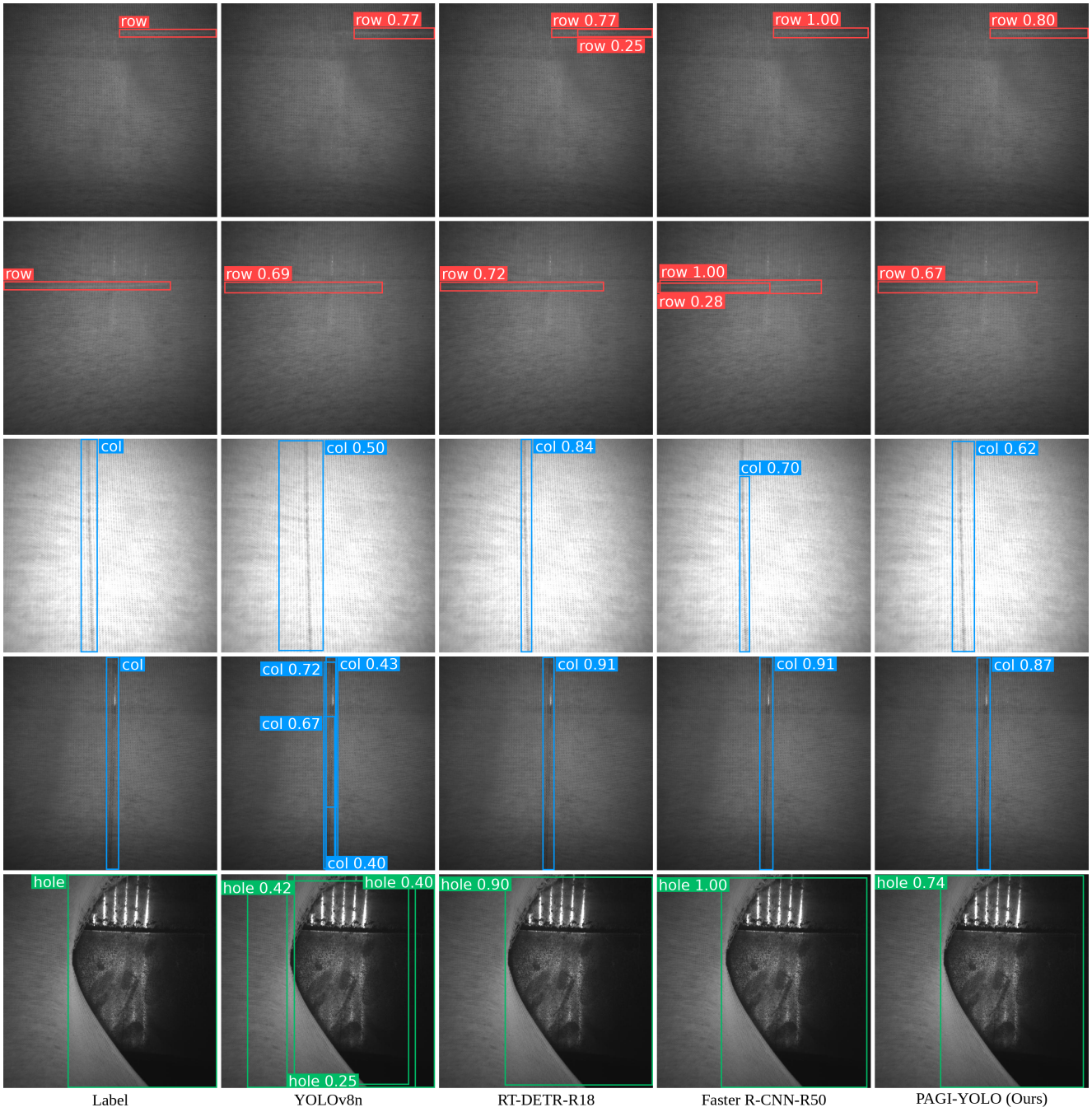


Fig. 6. Qualitative comparison of detection results on selected test images. From left to right: ground-truth annotations (Label), YOLOv8n, RT-DETR-R18, Faster R-CNN-R50, and MCGI-YOLO (Ours).

exceeding the 10 FPS design target. Future work will evaluate quantization-induced accuracy changes, end-to-end stopping performance, and false-stop frequency across fabric textures and operating conditions.

REFERENCES

[1] Cotton Incorporated, "Knit basics," CottonWorks, [Online]. Available: https://cottonworks.com/learning-hub/knitting/knit-basics/. Accessed: Oct. 8, 2026.

[2] The Woolmark Company, "Circular knit technology," [Online]. Available: https://www.woolmark.com/industry/product-development/processing-innovations/circular-knitting-technology/. Accessed: Oct. 8, 2026.

[3] Cotton Incorporated, "Single and double knits," CottonWorks, [Online]. Available: https://cottonworks.com/learning-hub/knitting/single-and-double-knits/. Accessed: Oct. 8, 2026.

[4] International Textile Manufacturers Federation, "Global textile machinery shipments shrunk in 2025, except for spinning," Jul. 2026, [Online]. Available: https://www.itmf.org/images/dl/press-releases/2026/2026-07-24-ITMSS-48-2025.pdf. Accessed: Oct. 10, 2026.

[5] United Nations Environment Programme, "Transforming the textile industry: Learning to manage waste, better," United Nations Environment Programme, Tech. Rep., Oct. 2025, [Online]. Available: https://wedocs.unep.org/handle/20.500.11822/48707. Accessed: Oct. 14, 2026.

[6] S. Takeuchi, K. Nishioka, H. Uematsu, and S. Tanoue, "Research into development of the defect detection system for knitted fabric produced by the circular knitting machines by image analysis," *Journal of Textile Engineering*, vol. 64, DOI 10.4188/jte.64.45, no. 2, pp. 45–49, 2018.

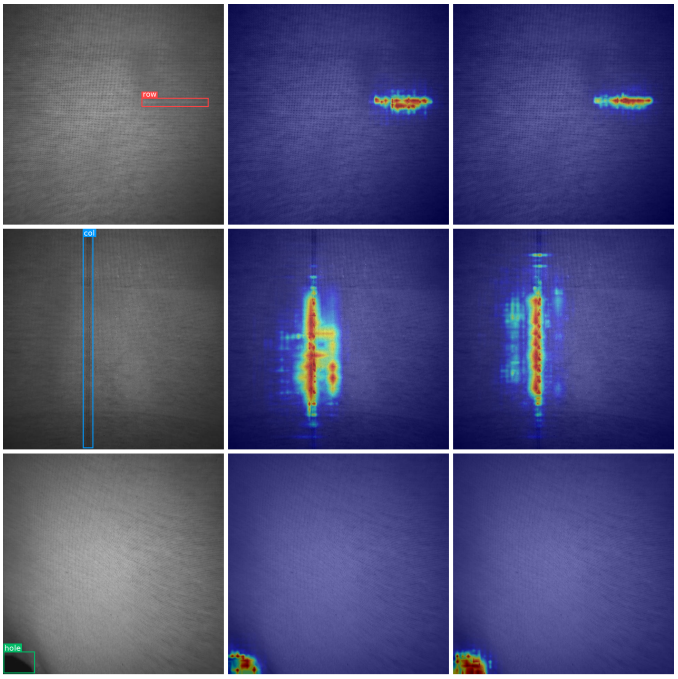


Fig. 7. Feature visualization of YOLOv8n and MCGI-YOLO. From left to right: ground-truth annotations (Label), YOLOv8n, and MCGI-YOLO.

[7] M. M. K. Akter, U. N. Haq, M. M. Islam, and M. A. Uddin, "Textile-apparel manufacturing and material waste management in the circular economy: A conceptual model to achieve sustainable development goal (SDG) 12 for bangladesh," *Cleaner Environmental Systems*, vol. 4, DOI 10.1016/j.cesys.2022.100070, p. 100070, 2022.

[8] Z. Xiang, Y. Shen, M. Ma, and M. Qian, "HookNet: Efficient multiscale context aggregation for high-accuracy detection of fabric defects," *IEEE Trans. Instrum. Meas.*, vol. 72, DOI 10.1109/TIM.2023.3280519, pp. 1–11, 2023, art. no. 5016311.

[9] S. Song, J. Jing, Y. Huang, and M. Shi, "EfficientDet for fabric defect detection based on edge computing," *J. Eng. Fibers Fabrics*, vol. 16, DOI 10.1177/15589250211008346, pp. 1–13, 2021.

[10] R. Machado, L. A. M. Barros, V. Vieira, F. D. da Silva, H. Costa, and V. Carvalho, "Textile defect detection using artificial intelligence and computer vision—a preliminary deep learning approach," *Electronics*, vol. 14, DOI 10.3390/electronics14183692, no. 18, p. 3692, 2025.

[11] T. Yan, Y. Wu, Y. Shi, C. Lin, and L. Ji, "Development and application of an AI visual defect detection system for warp-knitted lace based on 5G+ technology," *Information*, vol. 17, DOI 10.3390/info17070623, no. 7, p. 623, 2026.

[12] Y. Wang, H. Yu, B. Guo, H. Shi, and Z. Yu, "Research on real-time detection system of rail surface defects based on deep learning," *IEEE Sensors J.*, vol. 24, DOI 10.1109/JSEN.2024.3402730, no. 13, pp. 21 157–21 167, 2024.

[13] D. Li, Y. Lu, Q. Gao, X. Li, X. Yu, and Y. Song, "LiteYOLO-ID: A lightweight object detection network for insulator defect detection," *IEEE Trans. Instrum. Meas.*, vol. 73, DOI 10.1109/TIM.2024.3418082, pp. 1–12, 2024, art. no. 5023812.

[14] B. Zhao, Y. Wang, H. Zhang, J. Liu, and J. Zhang, "KAM-Net: Kilobyte-scale ultralightweight attention-based network for glass defect detection with algorithm/hardware co-design," *IEEE Trans. Instrum. Meas.*, vol. 74, DOI 10.1109/TIM.2025.3593602, pp. 1–10, 2025, art. no. 5041310.

[15] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 779–788, 2016.

[16] H. Yang, J. Jing, Z. Wang, Y. Huang, and S. Song, "YOLOV4-TinyS: A new convolutional neural architecture for real-time detection of fabric defects in edge devices," *Textile Research Journal*, vol. 94, DOI 10.1177/00405175231202035, no. 1–2, pp. 49–68, 2024.

[17] X. Kang and J. Li, "AYOLOv7-tiny: Towards efficient defect detection in solid color circular weft fabric," *Textile Research Journal*, vol. 94, DOI 10.1177/00405175231205898, no. 1–2, pp. 225–245, 2024.

[18] C. Chen, Q. Zhou, S. Li, D. Luo, and G. Tan, "Fabric defect detection algorithm based on improved YOLOv8," *Textile Research Journal*, vol. 95, DOI 10.1177/00405175241261092, no. 3–4, pp. 235–251, 2025.

[19] K. Yu, W. Lyu, X. Yu, Q. Guo, W. Xu, and L. Zhang, "FA-YOLO: A high-precision and efficient method for fabric defect detection in textile industry," *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. E107.A, DOI 10.1587/transfun.2023EAP1030, no. 6, pp. 890–898, 2024.

[20] Z. Zhang, T. Huang, W. Zhang, Y. Yang, and Q. Yu, "DGHR-YOLO: Fabric defect detection based on high-level screening-feature pyramid networks and deformable convolution," *Nondestructive Testing and Evaluation*, vol. 40, DOI 10.1080/10589759.2025.2451768, no. 12, pp. 5919–5944, 2025.

[21] B. Su, H. Chen, and Z. Zhou, "BAF-Detector: An efficient CNN-based detector for photovoltaic cell defect detection," *IEEE Trans. Ind. Electron.*, vol. 69, DOI 10.1109/TIE.2021.3070507, no. 3, pp. 3161–3171, 2022.

[22] X. Shen, J. Liu, L. Jiang, X. Liu, and H. Zhang, "A novel weld defect detection method for intelligent magnetic flux leakage detection system via contextual relation network," *IEEE Trans. Ind. Electron.*, vol. 71, DOI 10.1109/TIE.2023.3294578, no. 6, pp. 6304–6314, 2024.

[23] J. Liu, Z. Wen, X. Shen, F. Zuo, L. Jiang, and H. Zhang, "Online pipeline weld defect detection for magnetic flux leakage inspection system via lightweight rotated network," *IEEE Trans. Ind. Electron.*, vol. 72, DOI 10.1109/TIE.2024.3503635, no. 7, pp. 7573–7584, 2025.

[24] THine Electronics, Inc., "THCV241A: MIPI CSI-2 to V-by-One HS serializer," [Online]. Available: https://www.thine.co.jp/en/products/detail/THCV241A.html. Accessed: Oct. 16, 2026.

[25] THine Electronics, Inc., "THCV242A: V-by-One HS to MIPI CSI-2 deserializer," [Online]. Available: https://www.thine.co.jp/en/products/detail/THCV242A.html. Accessed: Oct. 16, 2026.

[26] Rockchip Electronics, "RK3566 brief datasheet," [Online]. Available: https://www.rock-chips.com/uploads/pdf/2022.8.26/191/RK3566%20Brief%20Datasheet.pdf. Accessed: Oct. 15, 2026.

[27] Ultralytics, "Ultralytics yolo," [Online]. Available: https://github.com/ultralytics/ultralytics, 2023, accessed: 2026-09-24.

[28] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, DOI 10.1109/CVPR.2017.106, pp. 2117–2125, 2017.

[29] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, DOI 10.1109/CVPR.2018.00913, pp. 8759–8768, 2018.

[30] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 13 713–13 722, 2021. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Hou_Coordinate_Attention_for_Efficient_Mobile_Network_Design_CVPR_2021_paper.html

[31] R. K. Srivastava, K. Greff, and J. Schmidhuber, "Training very deep networks," in *Advances in Neural Information Processing Systems*, vol. 28, pp. 2377–2385, 2015. [Online]. Available: https://proceedings.neurips.cc/paper/2015/hash/215a71a12769b056c3c32e7299f1c5ed-Abstract.html

[32] W. Lv *et al.*, "RT-DETR: Official repository with RT-DETRv2 implementations," [Online]. Available: https://github.com/lyuwenyu/RT-DETR, accessed: Oct. 20, 2026.

[33] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. Eur. Conf. Comput. Vis.*, DOI 10.1007/978-3-031-72751-1_1, pp. 1–21, 2025.

[34] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," in *Adv. Neural Inf. Process. Syst.*, vol. 37, DOI 10.52202/079017-3429, pp. 107 984–108 011, 2024.

[35] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," in *Adv. Neural Inf. Process. Syst.*, vol. 38, DOI 10.52202/085713-2627, 2025.

[36] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, DOI 10.1109/CVPR52733.2024.01605, pp. 16 965–16 974, 2024.

[37] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, pp. 213–229, 2020.

[38] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Representations*, 2021.

[39] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," in *Adv. Neural Inf. Process. Syst.*, vol. 28, 2015.
[40] Rockchip, "Rknn toolkit2," [Online]. Available: https://github.com/airockchip/rknn-toolkit2, 2026, accessed: 2026-09-27.

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